Theta oscillator

4 Hz, autonomous

$$\ddot{x}_\theta = -\omega_\theta^2 x_\theta - G_{\theta,0}\,\dot{x}_\theta + \sigma\xi_\theta$$

$$v_\theta = \dot{x}_\theta$$

$v_\theta(t)$

Theta-modulated friction

$$G_\gamma(t) = G_{\gamma,0} - g_{vv}\,v_\theta(t)$$

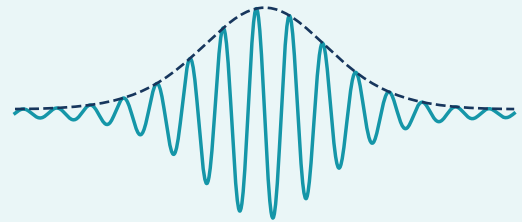
$v_\theta > 0$: less friction, gamma gains energy

$v_\theta < 0$: more friction, gamma loses energy

$G_\gamma(t)$

Gamma oscillator

60 Hz



envelope $\propto e^{g_{vv}x_\theta/2}$, peak at theta peak

Gamma energy balance

$$\mathrm{d}E_\gamma = \left\{-[G_{\gamma,0} - g_{vv}\,v_\theta(t)]v_\gamma^2 + \sigma^2/2\right\}\mathrm{d}t + \sigma v_\gamma\,\mathrm{d}W_\gamma$$